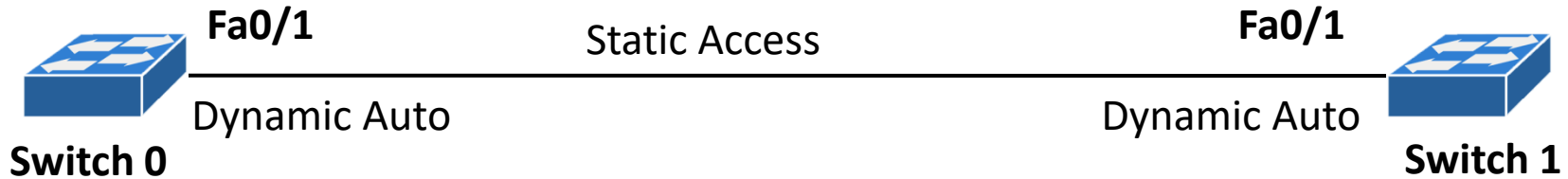


DTP LAB

Let us Assume we have two switch Switch 0 and Switch 1:

By Default interface of both Switches are in Dynamic Auto Mode
And combination of these two is still Static Access called
Operational mode



You can Verify this via command in privilege mode:

Switch0#show interfaces fa0/1 switchport

Command to Verify interface:

Switch0#show interfaces fa0/1 switchport

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Switch#show int fa0/1 SW
Switch#show int fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Ctrl+F6 to exit CLI focus
Copy Paste
```

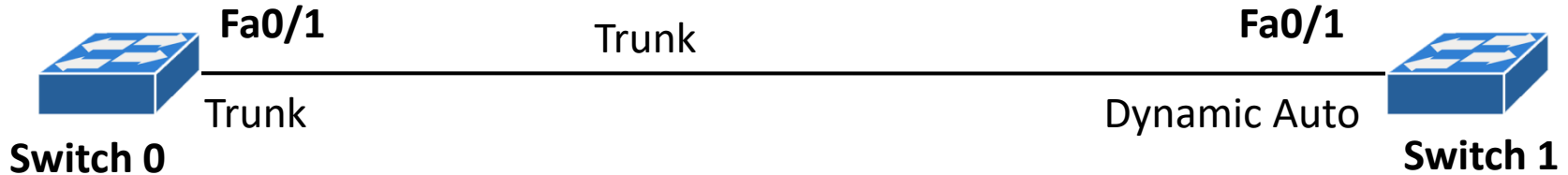
Interface Mode:
Administrative Mode
Is Dynamic Auto

Link Mode:
Operational Mode
Is Static Access
Because
Dynamic Auto +
Dynamic Auto =
Static Access

On Switch 0: Change the Mode to Trunk on One Side
The Link or Operational Mode will be changed to Trunk

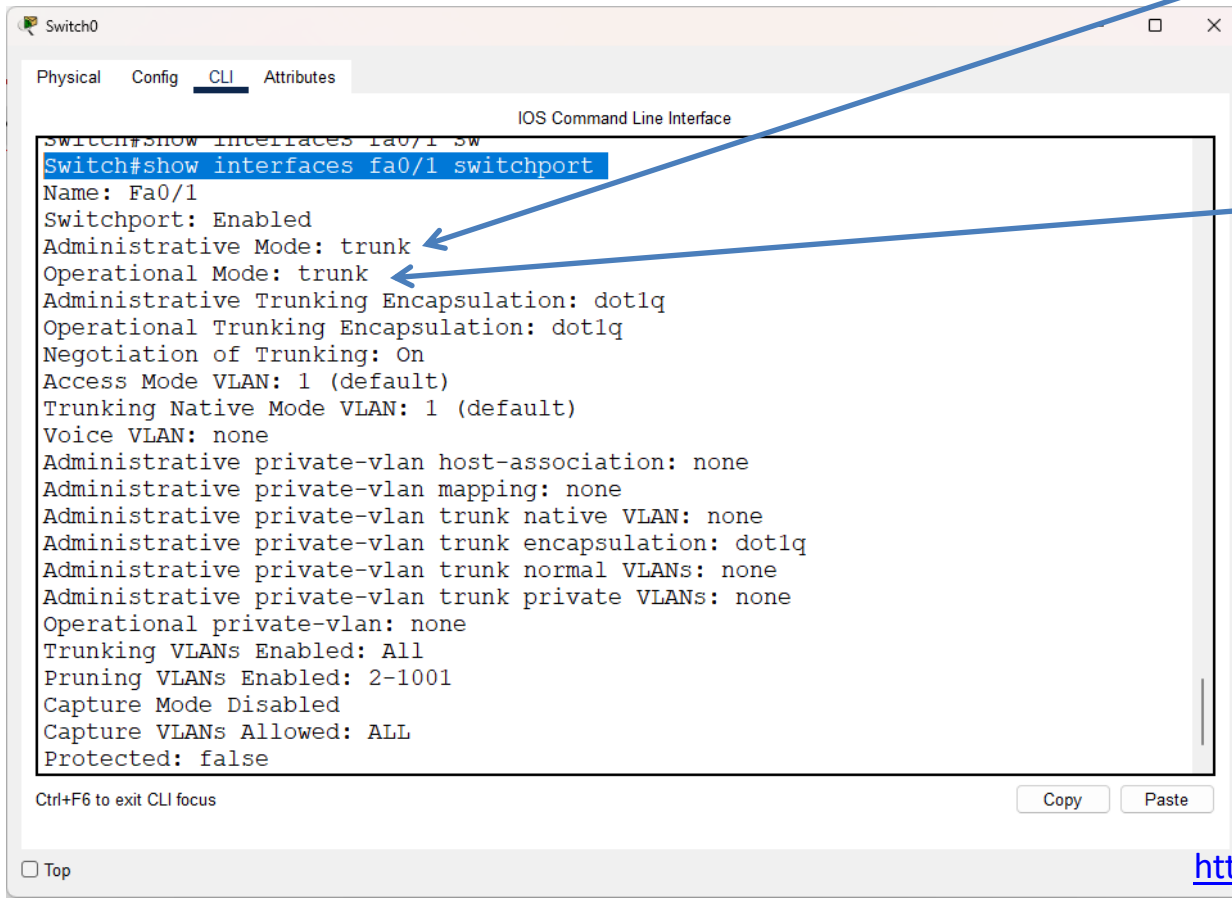
Switch0(Config)#Int fa0/1

*Switch0(config-if)#**Switchport Mode Trunk***



Verify the Trunk via Switchport:

Switch0#show interfaces fa0/1 switchport



```
Switch0#show interfaces fa0/1 switchport
Switch#show interfaces fa0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
```

Interface Mode:
Administrative Mode
Mode is Trunk

Link Mode:
Operational Mode
Is Static Access
Because
Trunk + Dynamic
Auto = Trunk

Verify Trunk:

Switch#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Fa0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Fa0/1	1-1005

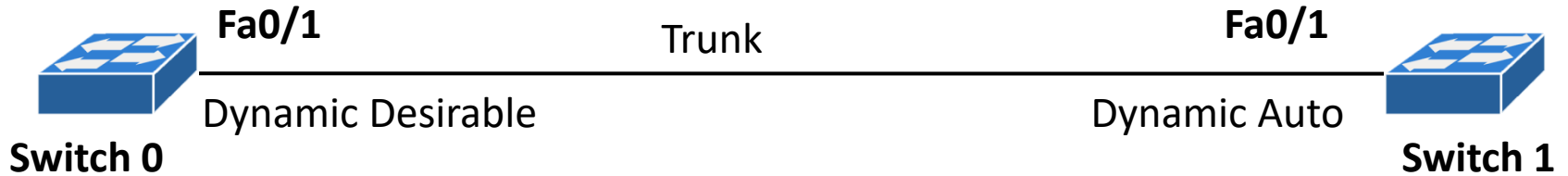
Port	Vlans allowed and active in management domain
Fa0/1	1

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/1	none

On Switch 0: Change the Mode to Dynamic Desirable on One Side
The Link or Operational Mode will be changed to Trunk

Switch(Config)#Int fa0/1

Switch(config-if)#Switchport Mode Dynamic Desirable



Switch0#show interfaces fa0/1 switchport

Administrative Mode: dynamic desirable

Operational Mode: trunk

Possible Combination of DTP:

Administrative Mode	Administrative Mode	Operational Mode
Access	Access	Static Access
Access	Trunk	Static Access
Access	Dynamic Auto	Static Access
Access	Dynamic Desirable	Static Access
Dynamic Auto	Dynamic Auto	Static Access
Trunk	Trunk	Trunk
Trunk	Dynamic Auto	Trunk
Trunk	Dynamic Desirable	Trunk
Dynamic Auto	Dynamic Desirable	Trunk
Dynamic Desirable	Dynamic Desirable	Trunk